The University of Lahore, CS & IT Department

Object Oriented Programming

Lab Assignment # 03
Part - 3: Hotel Management System

Start Date: 11/05/2025 | **Total Marks: 20**

Due Date: 18/05/2025 | **Program:** BSCS

Instructions

1. Understanding the problems is part of the assignment. So, no query, please.
2. You will get zero marks if found any type of plagiarism or AI generated code.
3. No submission after due date.
4. Submit the .cpp file containing your code.

Q1. Booking Request Management using Queue

1. Objective

Develop a Booking Request Management System using the Queue data structure. The system must handle both regular and high-priority booking requests, ensuring fairness (FIFO) for regular customers and urgency for priority cases.

2. Scenario: Galaxy Hotel – Booking Desk Manager

The Galaxy Hotel has a bustling reception desk receiving dozens of booking requests every day. The hotel has designed a digital request queue manager to ensure bookings are handled efficiently, in proper order. However, there are two classes of bookings:

- Regular Requests – added to a general queue and processed in FIFO batches.
- High Priority Requests – such as VIP or urgent cases, these must be processed before regular ones.

You've been tasked with building this booking request handler using Queues.

3. System Overview

Each booking request contains:

- Customer Name
- Room Type Requested: Single, Double, or Suite
- Number of Nights
- Priority: Regular / High
- Date of Request

Two queues:

- High Priority Queue
- Regular Request Queue
- Once 10 regular requests are queued, a batch is processed.
- High priority requests are always processed first.

4. **Required Functionalities**

   4.1. Add Booking Request
   - User inputs request details.
   - Based on priority, request is enqueued into the appropriate queue.

   4.2. Process Booking Requests
   - First check if any high priority requests exist — process them first.
   - Then process up to 10 regular requests in FIFO order.
   - Upon successful processing:
     - Assign room (use room manager).
     - Display booking success.

   4.3. Display Queue Status
   - Show number of requests in each queue.
   - List all pending requests.

   4.4. Check Queue Empty
   - Display message if either queue is empty.

   4.5. Display expected wait time based on queue size.

   4.6. Cancel a specific booking request from the queue.

5. **Program Requirements**
   - Use your own custom queue implementation using arrays or linked lists.
   - Do not use STL/Java libraries.
   - Menu-driven program with clear instructions.
   - Integrate all three parts and submit the final version.